Residual intersection properties for the Mathieu group of degree twenty-three

Human Compression project: bounded application

We apply the complete residue theorem in *Shadows and residues in finite coset geometries* and the sparse right-coset proof in *Shadows of the Hall–Janko geometries* to the forty-four previously classified locally two-transitive M_{23} geometries of ranks three through five. Both complete common proofs and the accepted phase-seven complete census are retained mathematical premises. No subgroup census is repeated.

1 The exact input population

The fixed supplementary archive for Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS J. Comput. Math. 7 (2004), 284–299, contains 13, 21 and 10 active families in `m23.rk3`, `m23.rk4` and `m23.rk5`. In these three files every construction entry remains active, in construction order. The literal archive parser verifies this fact; it is not inferred from a printed table total.

Let R_i denote entry i of the complete accepted phase-seven witness file. An actual ambient isomorphism identifies the source group with the retained simple group of order 10200960. Successive subgroup transporters under automorphisms, stabilizing every already placed subgroup, compare each entire parabolic family with every retained family of the same rank, allowing permutations of types. The complete matching proof is retained from the Hall–Janko adapter. Subgroup orders only screen candidates; actual subgroup equality establishes a match.

There is exactly one match for every archive entry, and no two entries match the same retained class. The matched sets are precisely

$$R_5, \ldots, R_{17}; \qquad R_{18}, \ldots, R_{38}; \qquad R_{39}, \ldots, R_{48}$$

in ranks three, four and five. Thus there are no omitted or duplicated classes in this boundary. The full 44-row correspondence is supplied. These retained identifiers are not asserted to be the historical names in the source's positive-geometry tables.

The earlier complete census supplies flag transitivity, firmness, residual connectivity, residual weak primitivity, local two-transitivity and $(\mathrm{IP})_2$ under its stated automorphism/type equivalence. An ambient isomorphism taking the whole parabolic family to another induces an incidence isomorphism on cosets and all residues. It therefore transfers all these predicates. The present run freshly checks flag extensions, residual generation, panel sizes, panel two-transitivity and all shadow decisions. Residual weak primitivity is explicitly retained from the complete census; no new primitivity computation is claimed.

2 All residual intersection decisions

Theorem 1. *The complete rank-specific counts are as follows. A triple means IP, weak IP alone, and failure of weak IP, respectively.*

Input rank	*number*	*rank three*	*rank four*	*rank five*	*full IP/WIP*
3	*13*	$(5,1,7)$	—	—	$(5,1,7)$
4	*21*	$(5,2,14)$	$(3,1,17)$	—	$(3,1,17)$
5	*10*	$(2,1,7)$	$(1,1,8)$	$(1,0,9)$	$(1,0,9)$

Every input satisfies $(\mathrm{IP})_2$.

Proof. For each proper fixed-type set J, the stabilizer H_J with parabolics $H_J \cap H_i$ is the base residue. Flag transitivity identifies every residue of that type with it, and $[G : H_J]$ gives the number of its translates. The common theorem reduces each directional shadow decision to the complete double-coset and flag-witness search. We run the exact accepted sparse algorithm, changing only its descriptive retained-census field. Nonempty shadows use canonical right-coset representatives; the empty flag is the whole universe and is handled by the proved cardinality criterion. Neither trivial panels nor digonal rank-two residues are silently excluded.

There are 716 base residues including panels, of which 543 have rank at least two. They give 1455 directional decisions. Every negative direction has an actual normalizing permutation and failed intersection after the complete witness search. Conjoining target directions and base residues of rank k gives $(\mathrm{IP})_k$. For $(\mathrm{WIP})_k$ a successful target may be chosen separately for each residue. Full IP and weak IP conjoin their respective rank-specific predicates for ranks two through the input rank. Panels are automatic and rank zero is excluded. Applying these exact quantifiers to all saved decisions proves the table. □

The full positive and weak-only classes are

rank	full IP	full weak IP alone
3	$9, 12, 13, 15, 17$	6
4	$20, 32, 37$	27
5	43	$\varnothing$.

All nine aggregate rows agree with the M_{23} rows of source Tables 1, 3, 5 and 7. The distinct lists in Tables 2, 4, 6 and 8 use historical classification names. Matching totals and truth profiles do not prove that label correspondence, which remains separately pending.

The published algorithm and reported source results belong to Jacobs and Leemans. The full six-page residue proof, full three-page sparse-adapter proof, and complete earlier census with its whole witness file remain paid dependencies. GAP 4.15.1 exact group, automorphism and coset routines are trusted. There is no new Lean, question, whole-original admission or numerical factor. Primary Section 6 contains executable Magma, so any future program-cost comparison must respect a matched source boundary.